UR5

6-DOF Upgraded Edition

Ur5 arm is configured and enables better configuration and is useful

Key Features

- good
- better

Applications

- day to day life
- speed
- education

Performance

Max TCP speed	0.50 m/s
Max joint speed	275.0°/s
Mounting Options	tabletop
Noise Level	¡ 60 dB(A)
Orientational Accuracy	± 0.714 °
Positional Accuracy	± 0.438 mm

General Specification

Payload	0.3 kg (at max reach)
Reach	846 mm
Degrees of Freedom	6
Configuration	Articulated Manipulator
Repeatability	±2.303 mm
Footprint	250

*Payload includes end-effector weight. Reach defined as maximum distance from base to TCP.

Joint Specifications

Joint	Working Range	Max Speed	Peak Torque
Shoulder Pan Joint	$-360°$ to $360°$	180°/s	150.0 Nm
Shoulder Lift Joint	$-360°$ to $360°$	180°/s	150.0 Nm
Elbow Joint	$-180°$ to $180°$	180°/s	150.0 Nm
Wrist 1 Joint	$-360°$ to $360°$	180°/s	28.0 Nm
Wrist 2 Joint	$-360°$ to $360°$	180°/s	28.0 Nm
Wrist 3 Joint	$-360°$ to $360°$	180°/s	28.0 Nm

*Gripper provides additional actuation DOF. Torque values are derived from manufacturer datasheets.

Control System

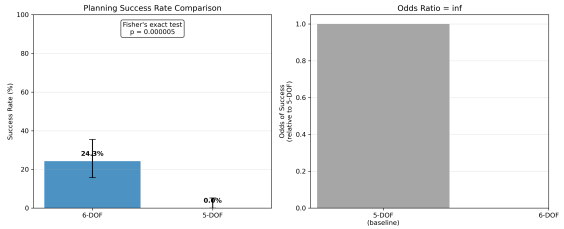
Supported OS	Ubuntu 22.04 / 24.04
Middleware	ROS 2 (Humble / Jazzy)
Motion Planning	MoveIt 2 (OMPL)
Controller	ros2_control
Simulation	Gazebo (Ignition)
Programming	Python, C++ APIs
Interface	USB / Serial (via controller)

Actuators

Actuator Type	motor
Motor Models	XM430-W350-T / XL430-W250-T
Control Mode	velocity
Communication	ttl
Feedback	Absolute Contactless Encoder

Electrical Specifications

Input voltage	13v
Power supply	12 V, 5 A adapter
Typical power	40w
Maximum power	100w (peak load)
Digital I/O	Available via external controller



Physical & Kinematics

Materials	High-strength polymer (PLA) + metal fasteners
Cable length	1m
Total Weight	≈ 0.8 kg
IP Rating	IP80
Operating Temp	0 °C to 45 °C
Humidity	5% to 85% (non-condensing)

Safety & Compliance

Safety rating	Uncertified (Research use only)
Architecture	Software-limited force/torque feedback (PL-b equivalent)
Protection	Configurable joint, velocity, and torque limits
E-Stop	Software-triggered (external hardware optional)

Known Limitations

- limitation

Documentation & Resources

GitHub Repository
Access source code, ROS 2 packages, setup guides, and updates.
github.com/KMTI-ROBOPARADIGM/omx-wrist-roll-original

Package Contents

- ur5 arm

Contact & Origin

Lab/Entity	**AutoCert Lab**
Contact	research@domain.edu
Location	India